

```
[cloudera@quickstart ~]$ ls
avro          eclipse          lib          rakeshdata1
cloudera-manager emp.java        Music        sparkjars_exec
cm_api.py      enterprise-deployment.json parcels        Templates
data1          express-deployment.json parquet_write Videos
Desktop        external_jars    part_dir     workspace
devices.json   external-unified Pictures      zeyo_tab.java
Documents      input.txt        Public
Downloads      kerberos         rakeshdata
```

```
[cloudera@quickstart ~]$ hadoop fs -ls
```

```
Found 15 items
```

```
drwxr-xr-x - cloudera cloudera    0 2020-05-23 23:09 avro_json_write
drwxr-xr-x - cloudera cloudera    0 2020-05-22 22:15 csv_dir
drwxr-xr-x - cloudera cloudera    0 2020-06-04 08:36 import_avro
drwxr-xr-x - cloudera cloudera    0 2020-05-23 22:56 json_avro_1
drwxr-xr-x - cloudera cloudera    0 2020-05-22 22:13 json_dir
drwxr-xr-x - cloudera cloudera    0 2020-05-23 22:39 json_orc
drwxr-xr-x - cloudera cloudera    0 2020-05-23 22:56 json_orc_1
drwxr-xr-x - cloudera cloudera    0 2020-05-23 22:38 json_parquet
drwxr-xr-x - cloudera cloudera    0 2020-05-23 22:56 json_parquet_1
drwxr-xr-x - cloudera cloudera    0 2020-05-22 22:11 orc_dir
drwxr-xr-x - cloudera cloudera    0 2020-05-22 22:14 parquet_dir
drwxr-xr-x - cloudera cloudera    0 2020-05-23 23:11 parquet_json_write
drwxr-xr-x - cloudera cloudera    0 2020-05-22 13:40 part_dir
drwxr-xr-x - cloudera cloudera    0 2020-05-22 14:01 part_dir2
drwxr-xr-x - cloudera cloudera    0 2020-06-04 09:04 zeyo_dir
```

```
[cloudera@quickstart ~]$ hadoop fs -mkdir anvin
```

```
[cloudera@quickstart ~]$ hadoop fs -ls
```

```
Found 16 items
```

```
drwxr-xr-x - cloudera cloudera    0 2026-03-12 03:59 anvin
drwxr-xr-x - cloudera cloudera    0 2020-05-23 23:09 avro_json_write
drwxr-xr-x - cloudera cloudera    0 2020-05-22 22:15 csv_dir
drwxr-xr-x - cloudera cloudera    0 2020-06-04 08:36 import_avro
drwxr-xr-x - cloudera cloudera    0 2020-05-23 22:56 json_avro_1
drwxr-xr-x - cloudera cloudera    0 2020-05-22 22:13 json_dir
drwxr-xr-x - cloudera cloudera    0 2020-05-23 22:39 json_orc
drwxr-xr-x - cloudera cloudera    0 2020-05-23 22:56 json_orc_1
drwxr-xr-x - cloudera cloudera    0 2020-05-23 22:38 json_parquet
drwxr-xr-x - cloudera cloudera    0 2020-05-23 22:56 json_parquet_1
drwxr-xr-x - cloudera cloudera    0 2020-05-22 22:11 orc_dir
drwxr-xr-x - cloudera cloudera    0 2020-05-22 22:14 parquet_dir
drwxr-xr-x - cloudera cloudera    0 2020-05-23 23:11 parquet_json_write
drwxr-xr-x - cloudera cloudera    0 2020-05-22 13:40 part_dir
drwxr-xr-x - cloudera cloudera    0 2020-05-22 14:01 part_dir2
drwxr-xr-x - cloudera cloudera    0 2020-06-04 09:04 zeyo_dir
```

```
[cloudera@quickstart ~]$ hadoop fs -put '/home/cloudera/Desktop/ok' /anvin
```

```
[cloudera@quickstart ~]$ hadoop fs -ls /anvin
```

```

-rw-r--r-- 1 cloudera supergroup      16 2026-03-12 04:00 /anvin
[cloudera@quickstart ~]$ hadoop fs -cat /anvin/ok
cat: `/anvin/ok': No such file or directory
[cloudera@quickstart ~]$ hadoop fs -cat anvin/ok
cat: `anvin/ok': No such file or directory
[cloudera@quickstart ~]$ hadoop fs -mkdir /anvin
mkdir: `/anvin': Is not a directory
[cloudera@quickstart ~]$ hadoop fs -cat anvin/ok
cat: `anvin/ok': No such file or directory
[cloudera@quickstart ~]$ hadoop fs -mkdir /anvi
[cloudera@quickstart ~]$ hadoop fs -put '/home/cloudera/Desktop/ok' /anvi
[cloudera@quickstart ~]$ hadoop fs -cat anvi/ok
cat: `anvi/ok': No such file or directory
[cloudera@quickstart ~]$ hadoop fs -put /home/cloudera/Desktop/ok /anvi/
put: `/anvi/ok': File exists
[cloudera@quickstart ~]$ hadoop fs -ls
Found 16 items
drwxr-xr-x - cloudera cloudera      0 2026-03-12 03:59 anvin
drwxr-xr-x - cloudera cloudera      0 2020-05-23 23:09 avro_json_write
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:15 csv_dir
drwxr-xr-x - cloudera cloudera      0 2020-06-04 08:36 import_avro
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:56 json_avro_1
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:13 json_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:39 json_orc
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:56 json_orc_1
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:38 json_parquet
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:56 json_parquet_1
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:11 orc_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:14 parquet_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-23 23:11 parquet_json_write
drwxr-xr-x - cloudera cloudera      0 2020-05-22 13:40 part_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-22 14:01 part_dir2
drwxr-xr-x - cloudera cloudera      0 2020-06-04 09:04 zeyo_dir
[cloudera@quickstart ~]$ hadoop fs -ls /anvi
Found 1 items
-rw-r--r-- 1 cloudera supergroup      16 2026-03-12 04:13 /anvi/ok
[cloudera@quickstart ~]$ hadoop fs -cat /anvi/ok
hi how are you

```

```

[cloudera@quickstart ~]$ hadoop fs -ls
Found 16 items
drwxr-xr-x - cloudera cloudera      0 2026-03-12 03:59 anvin
drwxr-xr-x - cloudera cloudera      0 2020-05-23 23:09 avro_json_write
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:15 csv_dir
drwxr-xr-x - cloudera cloudera      0 2020-06-04 08:36 import_avro
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:56 json_avro_1
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:13 json_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:39 json_orc

```

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drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:56 json_orc_1
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:38 json_parquet
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:56 json_parquet_1
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:11 orc_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:14 parquet_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-23 23:11 parquet_json_write
drwxr-xr-x - cloudera cloudera      0 2020-05-22 13:40 part_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-22 14:01 part_dir2
drwxr-xr-x - cloudera cloudera      0 2020-06-04 09:04 zeyo_dir
[cloudera@quickstart ~]$ hadoop fs -put /home/cloudera/Desktop/ok /anvin/
put: `/anvin': File exists
[cloudera@quickstart ~]$ hadoop fs -cat /anvin/ok
cat: `/anvin/ok': No such file or directory
[cloudera@quickstart ~]$ hadoop fs -ls /anvin
-rw-r--r--  1 cloudera supergroup      16 2026-03-12 04:00 /anvin
[cloudera@quickstart ~]$ hadoop fs -put /home/cloudera/Desktop/ok /anvin
put: `/anvin': File exists
[cloudera@quickstart ~]$ hadoop fs -mkdir /anvin1
[cloudera@quickstart ~]$ hadoop fs -put '/home/cloudera/Desktop/hello' /anvin1
[cloudera@quickstart ~]$ hadoop fs -cat /hello
cat: `/hello': No such file or directory
[cloudera@quickstart ~]$ hadoop fs -cat /anvin1/hello
hiiiiiiiiiiiiiiii
[cloudera@quickstart ~]$ hadoop fs -ls
Found 16 items
drwxr-xr-x - cloudera cloudera      0 2026-03-12 03:59 anvin
drwxr-xr-x - cloudera cloudera      0 2020-05-23 23:09 avro_json_write
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:15 csv_dir
drwxr-xr-x - cloudera cloudera      0 2020-06-04 08:36 import_avro
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:56 json_avro_1
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:13 json_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:39 json_orc
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:56 json_orc_1
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:38 json_parquet
drwxr-xr-x - cloudera cloudera      0 2020-05-23 22:56 json_parquet_1
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:11 orc_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-22 22:14 parquet_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-23 23:11 parquet_json_write
drwxr-xr-x - cloudera cloudera      0 2020-05-22 13:40 part_dir
drwxr-xr-x - cloudera cloudera      0 2020-05-22 14:01 part_dir2
drwxr-xr-x - cloudera cloudera      0 2020-06-04 09:04 zeyo_dir
[cloudera@quickstart ~]$ hadoop fs -ls /
Found 9 items
drwxr-xr-x - cloudera supergroup      0 2026-03-12 04:13 /anvi
-rw-r--r--  1 cloudera supergroup     16 2026-03-12 04:00 /anvin
drwxr-xr-x - cloudera supergroup      0 2026-03-12 04:24 /anvin1
drwxrwxrwx - hdfs    supergroup      0 2017-10-23 10:29 /benchmarks
drwxr-xr-x - hbase   supergroup      0 2026-03-11 21:04 /hbase

```

```

drwxr-xr-x - solr solr 0 2017-10-23 10:32 /solr
drwxrwxrwt - hdfs supergroup 0 2020-05-22 02:53 /tmp
drwxr-xr-x - hdfs supergroup 0 2020-05-22 02:53 /user
drwxr-xr-x - hdfs supergroup 0 2017-10-23 10:31 /var
[cloudera@quickstart ~]$ hadoop fs -rm /anvin1/hello
Deleted /anvin1/hello
[cloudera@quickstart ~]$ hadoop fs -rm /anvin/ok
rm: `/anvin/ok': No such file or directory
[cloudera@quickstart ~]$ hadoop fs -rm -r /anvin
Deleted /anvin
[cloudera@quickstart ~]$ hadoop fs -ls
Found 16 items
drwxr-xr-x - cloudera cloudera 0 2026-03-12 03:59 anvin
drwxr-xr-x - cloudera cloudera 0 2020-05-23 23:09 avro_json_write
drwxr-xr-x - cloudera cloudera 0 2020-05-22 22:15 csv_dir
drwxr-xr-x - cloudera cloudera 0 2020-06-04 08:36 import_avro
drwxr-xr-x - cloudera cloudera 0 2020-05-23 22:56 json_avro_1
drwxr-xr-x - cloudera cloudera 0 2020-05-22 22:13 json_dir
drwxr-xr-x - cloudera cloudera 0 2020-05-23 22:39 json_orc
drwxr-xr-x - cloudera cloudera 0 2020-05-23 22:56 json_orc_1
drwxr-xr-x - cloudera cloudera 0 2020-05-23 22:38 json_parquet
drwxr-xr-x - cloudera cloudera 0 2020-05-23 22:56 json_parquet_1
drwxr-xr-x - cloudera cloudera 0 2020-05-22 22:11 orc_dir
drwxr-xr-x - cloudera cloudera 0 2020-05-22 22:14 parquet_dir
drwxr-xr-x - cloudera cloudera 0 2020-05-23 23:11 parquet_json_write
drwxr-xr-x - cloudera cloudera 0 2020-05-22 13:40 part_dir
drwxr-xr-x - cloudera cloudera 0 2020-05-22 14:01 part_dir2
drwxr-xr-x - cloudera cloudera 0 2020-06-04 09:04 zeyo_dir
[cloudera@quickstart ~]$ hadoop fs -ls /
Found 8 items
drwxr-xr-x - cloudera supergroup 0 2026-03-12 04:13 /anvi
drwxr-xr-x - cloudera supergroup 0 2026-03-12 04:28 /anvin1
drwxrwxrwx - hdfs supergroup 0 2017-10-23 10:29 /benchmarks
drwxr-xr-x - hbase supergroup 0 2026-03-11 21:04 /hbase
drwxr-xr-x - solr solr 0 2017-10-23 10:32 /solr
drwxrwxrwt - hdfs supergroup 0 2020-05-22 02:53 /tmp
drwxr-xr-x - hdfs supergroup 0 2020-05-22 02:53 /user
drwxr-xr-x - hdfs supergroup 0 2017-10-23 10:31 /var
[cloudera@quickstart ~]$ hadoop fs -ls /anvi
Found 1 items
-rw-r--r-- 1 cloudera supergroup 16 2026-03-12 04:13 /anvi/ok
[cloudera@quickstart ~]$

```

MAP REDUCE

cd hadoop-0.20-mapreduce

hadoop jar hadoop-examples.jar = to see the examples inside

hadoop jar hadoop-examples.jar wordcount /anvi/vote /op1 -here we are going to do wordcount from then space filename of source then destination

hadoop jar hadoop-examples.jar grep /anvi/vote /op1 bjp

Here we use grep so source,destination,word to search for

What is mapreduce

Mapreduce is a distributed programming model used for processing large dataset across clusters of computers

It is part of the apache hadoop ecosystem and enables parallel data processing

Key Idea

Instead of processing data on a single machine,Hadoop splits data across multiple nodes and processes then in parallel.

Advantages

-handle big data(TBs/PBs)

-parallel processing

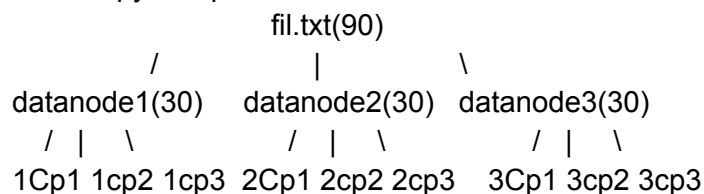
-fault tolerant

-works with commodity hardware"

How to make backup of datanode?

Every node will be having 3 copy by default which is the **replication factor(rf)=3**

It will copy 3 copies of datanode on 3 different datanode machine



Sudo jps - to check the datanodes

When we are running a program in mapreduce mainly 3 phase 1,map 2 ,reducer 3,main
In the word count program

2 .map phase

The map phase is the first stage of mapreduce

The mapper processes input data and convert it inot key-value pairs.

Steps:

1.input data is split into chunks

2.each chunk is processed by a mapper function

3,mapper outputs (key,value)pairs.

Exp:

Input:big data is powerful

Output:

(big,1)

(data,1)

(is,1)

(powerful,1)

3. Shuffle or sort phase

After map hadoop automatically perform shuffle and sort

Here groups all data by communicating with other nodes

Sort

The keys are sorted before sending to reducer

Program works as the contents first are maps->then sorted -> then reduced -> then output

4.reducer phase

Basics of yarn

Yarn yet another resource negotiator)is hadoop's resource management system.

It manages:

Cluster resource

Job scheduling

Application execution

Hadoop is a batch process

Olap-online analytical processing

Spark is real time oltp

Why yarn?

Before hadoop 2.x

Mapreduce handled both processing and resource management

With yarn

-resource management is separate

-support multiple processing engine

ex:.....:

Mapreduce

Spark

Flink

Tez

1. Resource Manager

Main cluster manager.

Responsibilities:

- Allocate cluster resources
 - Schedule applications
-

2. Node Manager

Runs on each worker node.

Responsibilities:

- Monitor resource usage
 - Manage containers
-

3. Application Master

Responsible for:

- Managing application execution
 - Communicating with Resource Manager
-

YARN Workflow

1. User submits job
 2. Resource Manager allocates resources
 3. Application Master starts
 4. Containers execute tasks
 5. Results are returned
-

- 1.resource manage
- 2.node manager
- 3,.application master

Yarn workflow

- 1.user submits job
- 2.resource manager allocates resources
- 3.application starts
- 4.containers execute tasks
- 5,result are returned